

# AI Agent for Smart Farming Advice – Question & Answers

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## 1. What is an AI Agent for Smart Farming Advice?

Answer:

An intelligent system that provides recommendations on crop selection, irrigation, fertilizers, pest management, and market insights.

## 2. What is Agentic AI?

Answer:

Agentic AI can reason, plan, retrieve information, and perform tasks autonomously to achieve goals.

## 3. How does the Smart Farming Agent help farmers?

Answer:

It provides personalized recommendations and answers farming-related queries.

## 4. What technologies are used?

Answer:

Langflow, IBM watsonx.ai, IBM Granite Models, RAG, Vector Databases, and Python.

## 5. What is RAG?

Answer:

Retrieval-Augmented Generation retrieves information from trusted agricultural documents and combines it with AI-generated responses.

## 6. What is the role of the Crop Advisory Agent?

Answer:

It recommends suitable crops, seeds, and cultivation practices.

## 7. What is the role of the Weather & Irrigation Agent?

Answer:

It provides irrigation schedules and weather-based recommendations.

### **8. What is the role of the Pest & Disease Agent?**

Answer:

It helps identify crop diseases and suggests preventive measures.

### **9. What is the role of the Market Insights Agent?**

Answer:

It provides crop prices and market trend information.

### **10. How does the system generate recommendations?**

Answer:

The AI model analyzes queries, retrieves knowledge, and generates responses.

### **11. What are the benefits of Smart Farming AI?**

Answer:

Improved productivity, better resource management, and higher profitability.

### **12. Can the system support sustainable farming?**

Answer:

Yes, it promotes efficient and environmentally friendly farming practices.

### **13. Why is Langflow used?**

Answer:

Langflow provides a visual workflow environment for AI agents.

### **14. What is IBM watsonx.ai?**

Answer:

IBM's enterprise AI platform for building generative AI applications.

### **15. What are IBM Granite Models?**

Answer:

IBM foundation models used for reasoning and content generation.

### **16. Can farmers ask questions in natural language?**

Answer:

Yes, through a conversational AI interface.

**17. What are the limitations of the system?**

Answer:

Dependence on data quality and availability of updated information.

**18. What is the future scope of the project?**

Answer:

IoT, drones, satellite imagery, and multilingual voice assistants.

**19. How does the project improve decision making?**

Answer:

By providing timely and data-driven recommendations.

**20. Why is this project important?**

Answer:

It bridges the gap between farmers and expert agricultural knowledge.